Department Meeting:

May 18, 58.

Topic: Dept 549 : Circuits Level & Standardization — Joe Sogno

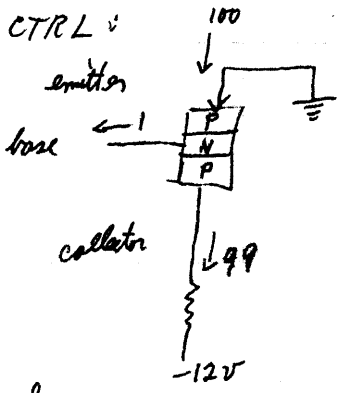
range from 024 → 7000.

to get: optimum cost - optimum performance.

Range of 7000 : Drift current mode circ. ← used some way in circ.

785 : alloy current mode circ. ↙ (non saturating)

~ 608 / 024 : CTRL complementary Transistor resistor logic.



conducts if P is pos.

← one mill controls 99 mils

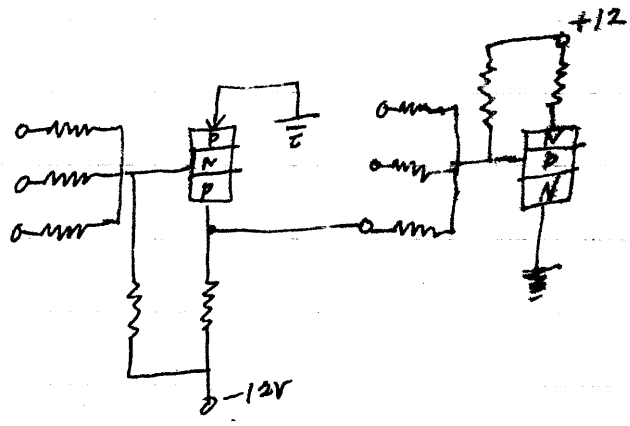
bank resistance very large 5×10^{-6} amps

100×10^{-3} amps.

Speed

Alloy : 0.2 - 0.4 μs

CTRL



can get

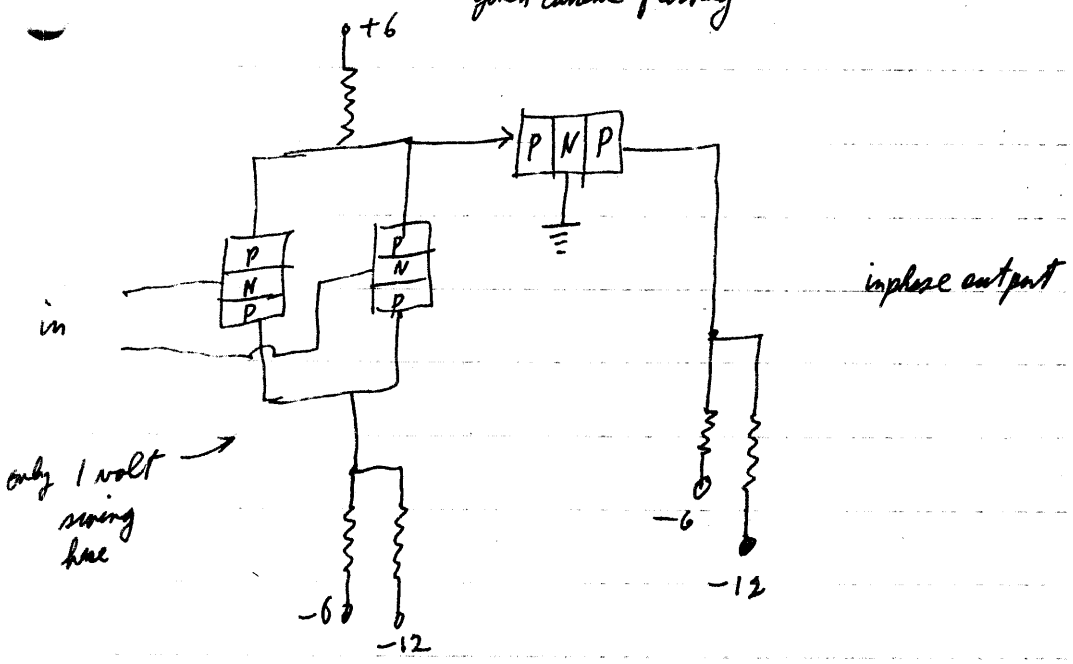
$Z = A \cdot B + C \cdot D \cdot X$

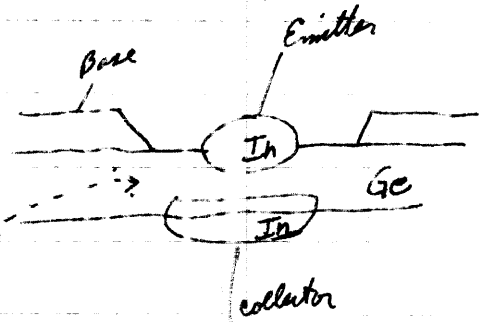
with 2 transistors.

This instead of 0.2 0.3

Permicinake relay : faster than wire-contact. more reliable ? used in 088, 305

will it go into production ? to be decided end of this year.





These Transistors are not saturated

when saturated carriers are emitted

lifetime 10-50 μs

- current continues to flow until these are all collected.
- also capacitance between collector + base are higher -

also α-cut-off freq. changes. (base width is effectively smaller)

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another possibility: use CTRL with shifts, about 2.1 usec speed.

Marginal checking

Power supplies ±4%, CTRL ±10%

drivers, relay drivers,

Manuals: 3 more out.

[Hot Spot, 122°F]

some Transistors can hold off 200V, 20-30 amps

GE Silicon Tra. 10 amps - 200-300 V holdoff.

[dividers; another possibility faster circ. for 785?

Reliability.

In 608: 8 transistors have failed — most due to whiskers coming out.
[illegible] 2 million Tr hrs between failures.
- saved "can" the circuits for no servicing

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core logic: big disadvantage - no gain bandwidth
expensive & not as fast.
a combination of core for logic with transistors for gain.

1952-3: transistorized 475 with point-contact Trs. — poor reliability.
Spr. 1953: IBM went to alloy junctions even though they were slower.
Machines were laid out. 607, 475, data transceiver. (604 was actually built in about 3 mos.
Signal swing: low voltages.

on 608: demanded that we have both PNP's & NPN's. competitors went along with only one type PNP, eg Philco.

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weak on Silicon Transistors: , TI, Philco, GE, Raytheon are all doing better than IBM.

IBM guesses 44¢ per Tr in 1960-61 (about $2.50 now.)
TI 83¢

Diode Logic: ? probably on its way out. — need large signal swings and high current to operate diodes.
- this increases temps on transistors & makes them fail,
— cost is not much lower

no. pluggable units in machine / no. of types = "Standardization Factor"

RTA Test assembly

DC pulse circuits

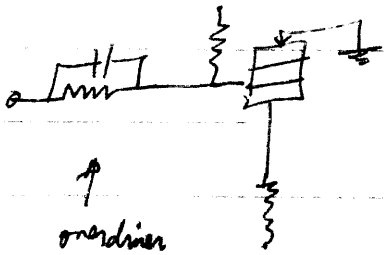
Delta Junction: Philco microalloy transistors.

9 to 15 nsec delay per stage

120 nsec delays due to pulse transformer logic, picked down string.

DCTL direct coupled XSTR logic

modified



can cause spikes due to partial charges. — go right thru C.